

Muhammad Aqeel

Curriculum Vitae

Dept. of Engineering for Innovation Medicine
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Research Interests

My research is centered on anomaly detection for industrial inspection, with a focus on unsupervised and zero-shot settings where labeled defect data is not available. Over time my work has naturally expanded toward generative models and vision-language approaches, driven by the need to handle data scarcity and improve generalization across unseen defect categories. Recently I am also exploring video understanding and multimodal summarization, though my main research remains in anomaly detection.

Employment

- 2026–Present **Postdoctoral Research Fellow**, *University of Verona*, Verona, Italy.
Jun–Nov 2025 **Visiting Researcher**, *University of Ljubljana*, Ljubljana, Slovenia.
2023–2025 **Teaching Assistant**, *University of Verona*, Verona, Italy.
2021–2022 **Research Assistant**, *Huazhong University of Science and Technology*, Wuhan, China.

Education

- 2022–2025 **PhD in Computer Science**, *University of Verona*, Italy.
Supervisors: Prof. Francesco Setti & Prof. Marco Cristani
2019–2021 **Master in Computer Science**, *Beijing University of Posts and Telecommunications*, China.
Supervisor: Prof. Xiaohong Liu
2013–2017 **Bachelor in Computer Science**, *PMAS Arid Agriculture University*, Rawalpindi, Pakistan.

Publications

Journal Articles

- 2025 **M. Aqeel & F. Setti**, *Semantic Parsing for Aspect Based Sentiment Analysis*, *IEEE Access*, Vol. 13, pp. 127322–127334.

Conference Proceedings

- 2026 **M. Aqeel, M. Nazir, U. Khan, M. Cristani & F. Setti**, *Anomaly-Aware Vision-Language Adapters for Zero-Shot Anomaly Detection*, *IEEE International Conference on Image Processing (ICIP)*.
2026 **M. Nazir, R. Zhang, M. Aqeel & F. Setti**, *Multimodal Abstractive Summarization of Instructional Videos with Vision-Language Models*, *International Conference on Pattern Recognition (ICPR)*.
2025 **M. Aqeel, S. Sharifi, M. Cristani & F. Setti**, *Towards Real Unsupervised Anomaly Detection via Confident Meta-Learning*, *IEEE/CVF International Conference on Computer Vision (ICCV)*.
2025 **M. Aqeel, S. Sharifi, M. Cristani & F. Setti**, *Robust Anomaly Detection in Industrial Environments via Meta-Learning*, *IEEE/CVF International Conference on Computer Vision (ICCV)*.

- 2025 **M. Aqeel, D. Skočaj, M. Cristani & F. Setti**, *A Contrastive Learning-Guided Confident Meta-Learning for Zero-Shot Anomaly Detection*, *IEEE/CVF International Conference on Computer Vision (ICCV)*.
- 2025 **M. Aqeel, F. Leonardi & F. Setti**, *ExDD: Explicit Dual Distribution Learning for Surface Defect Detection via Diffusion Synthesis*, *International Conference on Image Analysis and Processing (ICIAP)*, Springer.
- 2025 **M. Aqeel, K. D. Bellele & F. Setti**, *RoadFusion: Latent Diffusion Model for Pavement Defect Detection*, *International Conference on Image Analysis and Processing (ICIAP)*, Springer.
- 2025 **M. Aqeel, M. Nazir, Z. Ruan & F. Setti**, *Latent Space Synergy: Text-Guided Data Augmentation for Direct Diffusion Biomedical Segmentation*, *ICIAP Workshops*, Springer.
- 2025 **M. Nazir, M. Aqeel & F. Setti**, *Diffusion-Based Data Augmentation for Medical Image Segmentation*, *IEEE/CVF International Conference on Computer Vision (ICCV)*.
- 2025 **M. Aqeel, S. Sharifi, M. Cristani & F. Setti**, *Self-Supervised Iterative Refinement for Anomaly Detection in Industrial Quality Control*, *Int. Conf. on Computer Vision Theory and Applications (VISAPP)*, SCITEPRESS, pp. 173–183.
- 2024 **M. Aqeel, S. Sharifi, M. Cristani & F. Setti**, *Meta Learning-Driven Iterative Refinement for Robust Anomaly Detection in Industrial Inspection*, *European Conference on Computer Vision (ECCV)*, Springer, pp. 445–460.
- 2024 **M. Aqeel, S. Sharifi, M. Cristani & F. Setti**, *Self-supervised Learning for Robust Surface Defect Detection*, *Int. Conf. on Deep Learning Theory and Applications (DeLTA)*, Springer, pp. 164–177.
- 2024 **L. Capogrosso, F. Girella, F. Taioli, M. D. Chiara, M. Aqeel, F. Fummi, F. Setti & M. Cristani**, *Diffusion-Based Image Generation for In-Distribution Data Augmentation in Surface Defect Detection*, *Int. Conf. on Computer Vision Theory and Applications (VISAPP)*, SCITEPRESS, pp. 409–416.

Funding & Grants

- 2022–2025 **PNRR Funding for PhD**, *University of Verona*, Italy.
- 2025–2025 **Mobility Research Grant**, *University of Ljubljana*, Slovenia.
- 2019–2021 **Chinese Government Scholarship (CSC) for Master**, *Beijing University of Posts and Telecommunications*, China.

Awards & Honours

- 2025 Certificate of Oral Presentation, Workshop on Computer Vision and Generative Models for Medical Imaging, ICIAP 2025

Academic Service

Reviewer – Journals

IEEE Transactions on Multimedia
 Neural Computing and Applications
 Applied Soft Computing

Reviewer – Conferences

IEEE/CVF International Conference on Computer Vision (ICCV) 2025
 European Conference on Computer Vision (ECCV)
 IJCAI-ECAI
 Int. Conf. on Computer Vision Theory and Applications (VISAPP) 2025

Conferences & Presentations

- Sep 2025 **International Conference on Image Analysis and Processing (ICIAP)**, Rome, Italy.
Oral: *RoadFusion: Latent Diffusion Model for Pavement Defect Detection*
Poster: *ExDD: Explicit Dual Distribution Learning for Surface Defect Detection*
Oral: *Latent Space Synergy* – CVGMMI Workshop
Invited Talk – T4S Workshop (Traffic and Street Surveillance for Safety and Security)
- Sep–Oct 2024 **European Conference on Computer Vision (ECCV)**, Milan, Italy.
Oral: *Meta Learning-Driven Iterative Refinement for Robust Anomaly Detection* – VISION Workshop
- Feb–Mar 2025 **Int. Conf. on Computer Vision Theory and Applications (VISAPP)**, Porto, Portugal.
Oral: *Self-Supervised Iterative Refinement for Anomaly Detection in Industrial Quality Control*
- Feb 2024 **Int. Conf. on Computer Vision Theory and Applications (VISAPP)**, Rome, Italy.
Poster: *Diffusion-based Image Generation for In-distribution Data Augmentation in Surface Defect Detection*

Memberships

- Since 2025 Member, CVPL – Italian Association for Computer Vision, Pattern Recognition and Machine Learning

Teaching Assistantship

- 2023–2025 **Teaching Assistant**, University of Verona, Verona, Italy.
Computer Vision and Machine Learning, Dept. of Engineering for Innovation Medicine

Technical Skills

- Languages Python, C, C++, Java, MATLAB
Frameworks PyTorch, TensorFlow, Keras, Hugging Face
Tools Git, Docker, L^AT_EX, Anaconda, PyCharm, Visual Studio
Databases MySQL, Oracle, Neo4j
OS Linux, macOS, Windows

Languages

- Urdu / **Native**
Punjabi
English **Proficient (C2)**
Italian **Basic (A2)**
Chinese **Basic (A2)**

Referees

Prof. Francesco Setti

Associate Professor

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Prof. Marco Cristani

Professor

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Prof. Liu Xiao Hong

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